

Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Introduction to Packet Tracer and Network Topologies
Lab Number	02
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab Tasks

Statement Purpose:

Get acquainted with Packet Tracer Make some simple Packet Tracer scenarios.

Activity outcomes:

- Students will have gained the basic understanding of Packet Tracer to see “protocols in action”.
- Students will have developed basic understanding of digging deep into the network protocols.

Introduction to the Packet Tracer Interface.

Network Topology:

Network topology is the way devices are connected in a network. It defines how these components are connected and how data transfer between the network. There are two major categories of Network Topology i.e. Physical Network topology and Logical Network Topology.

Types of Network Topology:

Below mentioned are the types of Network Topology.

- [Point to Point Topology](#)
- [Mesh Topology](#)
- [Star Topology](#)
- [Bus Topology](#)
- [Ring Topology](#)
- [Tree Topology](#)
- [Hybrid Topology](#)

Post Lab Task:

Post Lab Task 1: Network Topologies – Advantages and Disadvantages

i. Mesh Topology:

Advantages:

- Redundancy ensures no single point of failure.
- High fault tolerance and reliability.
- Supports high traffic; dedicated links.

Disadvantages:

- Expensive to implement due to number of cables/ports.
- Complex configuration and maintenance.
- Not scalable for large networks.

ii. Star Topology:

Advantages:

- Easy to install and manage.
- Failure of one node does not affect the others.
- Centralized management through hub/switch.

Disadvantages:

- Failure of central hub/switch breaks the network.
- Requires more cable than bus topology.
- Cost increases with central device quality.

iii. Bus Topology:**Advantages:**

- Simple and inexpensive to install.
- Requires less cable than star or mesh.

Disadvantages:

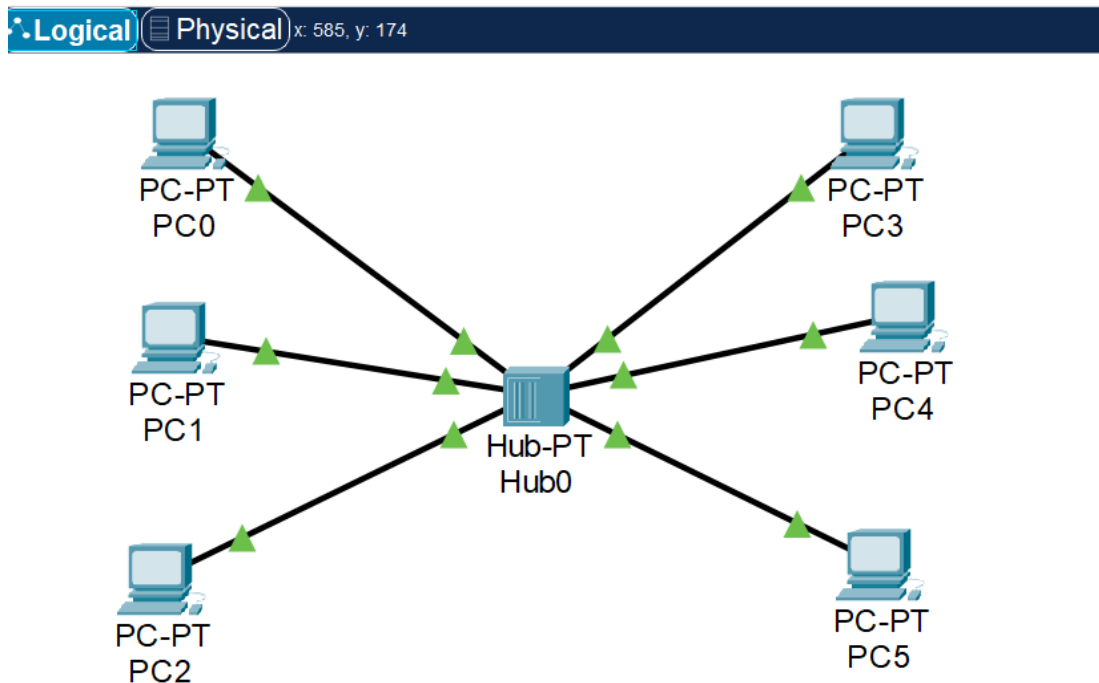
- Single point of failure (main cable).
- Limited cable length and number of stations.
- Difficult to troubleshoot and not scalable.

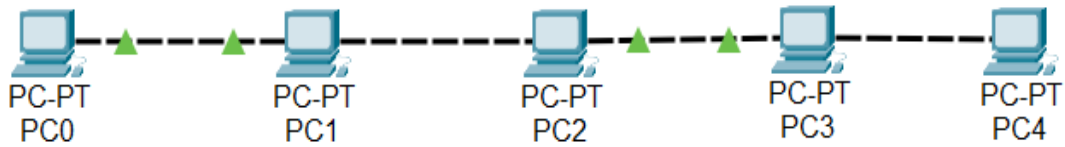
iv. Ring Topology:**Advantages:**

- Equal access to resources.
- Simple and orderly network access.

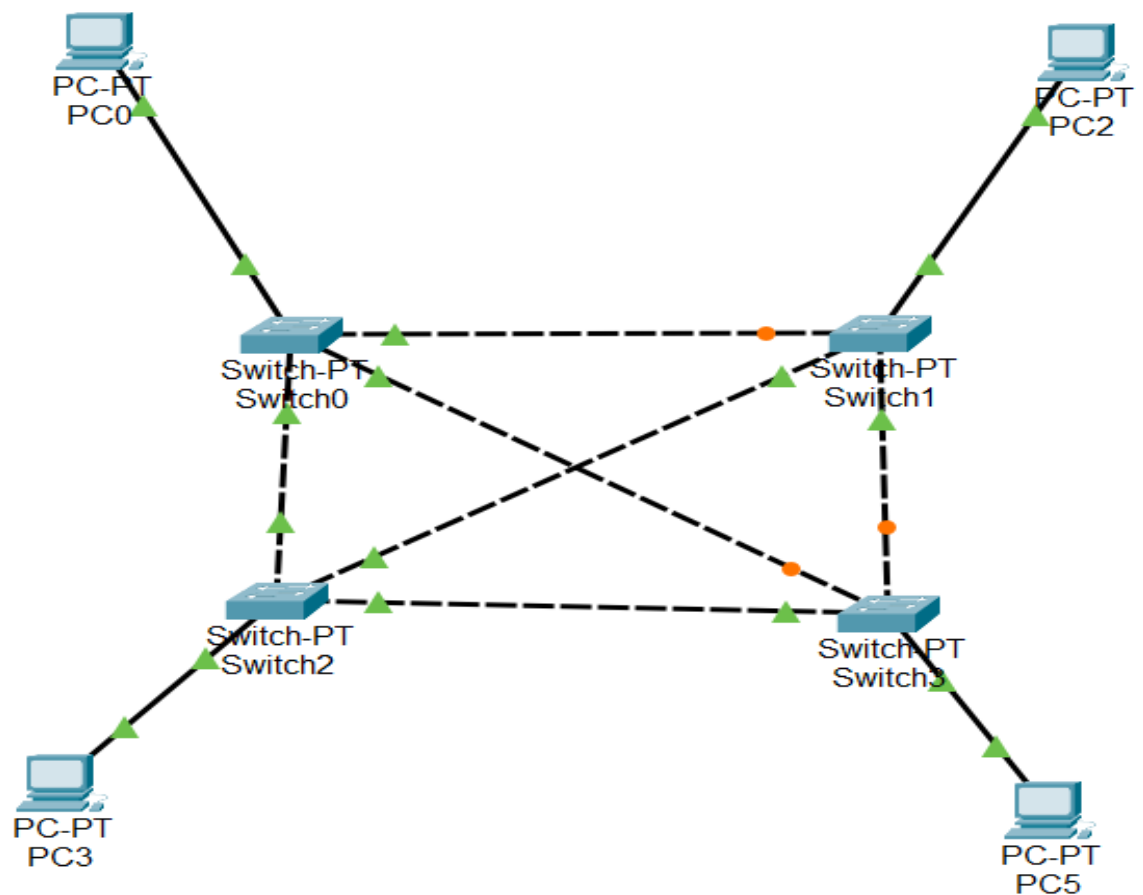
Disadvantages:

- Failure in any cable or device disrupts the whole network.
- Difficult to reconfigure or troubleshoot.
- Data must pass through all devices in the ring.

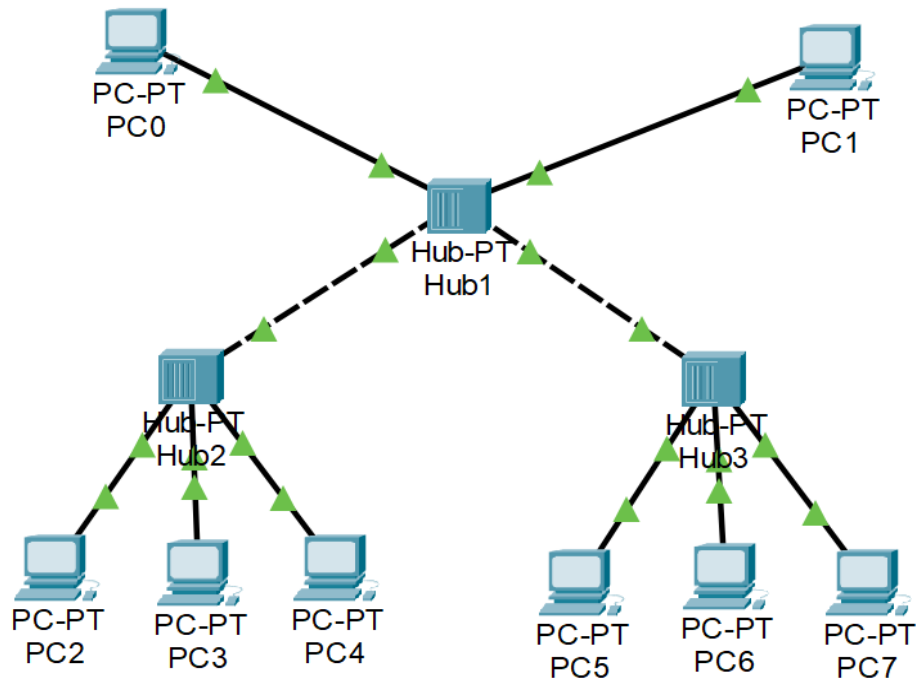
Post Lab Task 2: Topology Implementation in Packet Tracer:**➤ Star Topology:****➤ Bus Topology:**



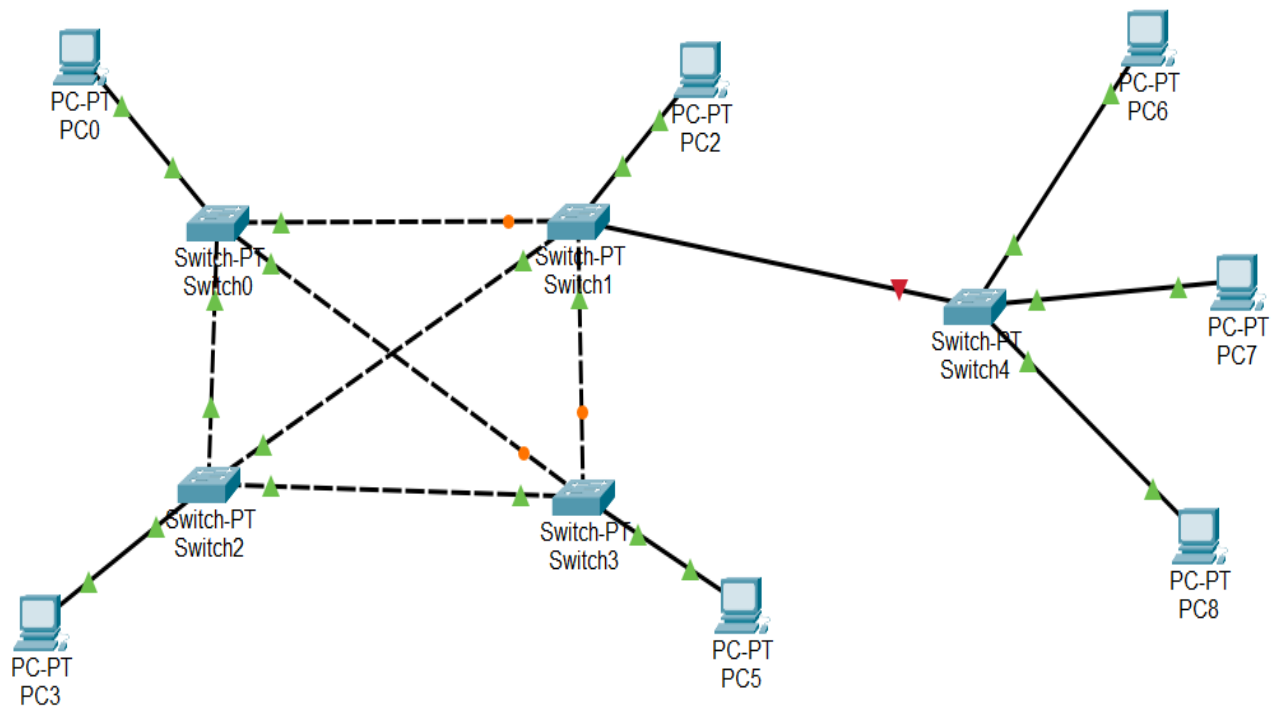
➤ **Mesh Topology:**



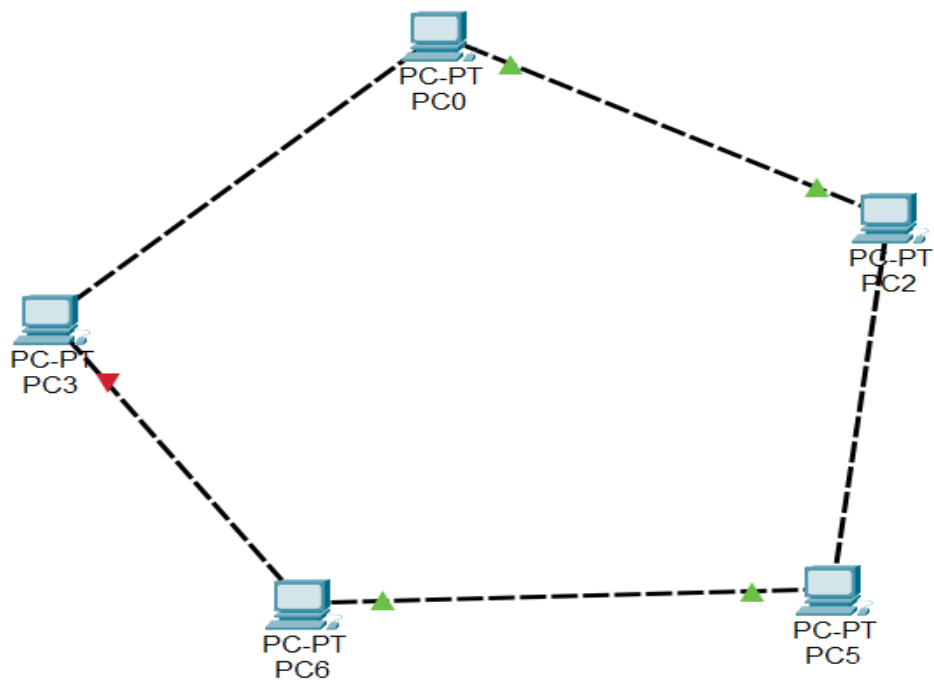
➤ **Tree Topology:**



➤ Hybrid Topology:



➤ Ring Topology:



Conclusion:

In conclusion, network topologies play a crucial role in determining the efficiency and reliability of a computer network. Each topology, whether it's bus, star, ring, mesh, or tree, offers unique benefits and potential drawbacks. By understanding these different arrangements, network designers can choose the most appropriate topology to meet the specific needs of their systems, ensuring optimal performance and connectivity.